

OWL Scientific Question

Topic: Ocean Worlds Exploration & Sampling [1, pp. 36–39]

Interest:

Ocean worlds like Europa represent the most compelling environments for studying life beyond Earth, and the most difficult for autonomous systems. What draws me to this topic is the engineering challenge that arise from the question: how do you keep a robot operational, navigating, and decision-making in a GPS-denied, high-radiation, cryogenic environment with no possibility of real-time human intervention? The decadal survey speaks to the real physical needs here: Jupiter’s Galilean moons reveal “the fundamental importance of gravitational tides in generating internal heat,” with those tides actively “melting ice on Europa and Ganymede” [1, p. 37]. That subsurface ocean exists because of a process we cannot turn off, and any robotic system sent to study it must be designed around that reality. These are the exact conditions that highlight every weakness in current autonomy stacks.

Connection to Space Resources:

Accessing a subsurface ocean requires penetrating kilometers of ice, a process that may yield in-situ resources (liquid water, chemical energy gradients) critical to sustaining any long-duration mission. Understanding the composition of that ocean directly informs the feasibility of using local materials to support future human or robotic infrastructure in the outer solar system.

Relationship to Robotics:

Reaching and operating in a subsurface ocean demands a full robotic stack operating without outside reference: special robots for ice penetration, underwater vehicles for ocean navigation, and autonomous sampling systems capable of detecting and responding to scientifically interesting features. Every subsystem (perception, localization, path planning, controls) must function independently.

STMD Technology Shortfall

Topic: Multi-Agent Robotic Coordination and Interoperability for Cooperative Task Planning and Performance [2]

Interest:

Swarm robotics is the next step in advancing the capabilities of robots to complete high level missions. NASA’s own shortfall rankings confirm how urgent this is: multi-agent coordination received an integrated score of 4.7856 and ranked 139th among all civil space technology gaps [2], placing it right in the near-term priority tier for Autonomous Systems and Robotics. I am most interested in the version of it where each robot in the swarm is genuinely capable on its own, not a simple unit that only functions within the group (similar to how drone swarms work). Instead we’d have a system of robots with unique capabilities that all come together to complete a high-level task.

Connection to Space Resources:

Cooperative exploration and mapping by a distributed robotic fleet dramatically increases the spatial coverage and fault tolerance of resource missions. A swarm can survey a large asteroid surface, map subsurface ice deposits via distributed sensing, or maintain constant monitoring of a volatile-rich region far more efficiently than any single agent, with adversarial identification algorithms when individual units fail.

Relationship to Robotics:

This topic is a direct application of the consensus and coordination algorithms I have studied

for the last year. Deploying them in space introduces new constraints (limited bandwidth, high latency, radiation-induced faults, and no GPS) that stress-test the theoretical properties I've studied (convergence, resilience, scalability). Building swarm systems that consistently maintain coordination under these conditions allows me to further see the applications of this field.

NASA SBIR Research Topic

Topic: Flight Dynamics and Navigation Technologies [3] (Topic COMNAV.1.S26B)

Interest:

Communication delays and blackouts are not random edge cases in deep-space missions; they are the default operating condition. The SBIR solicitation states the gap directly: “NASA currently has limited navigational and guidance technologies that permit fully autonomous approach, proximity operations, distributed spacecraft operations, and landing without support from Earth-based resources” [3]. This shortfall interests me because it forces onboard autonomy to be the primary decision-making layer, eliminating the safety net of human-in-the-loop oversight. A robot that cannot function robustly without a communication link is not truly autonomous. Closing this gap is a requirement for any serious deep-space resource or exploration mission.

Connection to Space Resources:

Remote resource operations (prospecting, extraction, processing) cannot depend on continuous communication to Earth. Whether managing an ISRU plant on the lunar surface or conducting a sampling traverse on an asteroid, the system must detect anomalies, replan, and execute without waiting for human feedback. Autonomous operations capability is therefore a direct necessity for in-space resource utilization.

Relationship to Robotics:

My research in consensus algorithms and resilient multi-agent coordination already deals with how distributed systems reach agreement under adversarial conditions. Extending this to communication-constrained space contexts is a natural progression my PI has brought up: the same algorithms that allow a swarm of Crazyflies to maintain stability during packet loss should also apply to a team of planetary rovers operating through a communication outage.

References

- [1] National Academies of Sciences, Engineering, and Medicine, *Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023–2032*, The National Academies Press, Washington, D.C., 2022. [Online]. Available: <https://www.nationalacademies.org/read/26522>. [Accessed: Jun. 14, 2026].
- [2] NASA Space Technology Mission Directorate, “Civil Space Technology Shortfall Rankings,” NASA, Washington, D.C., Jul. 2024. [Online]. Available: <https://www.nasa.gov/wp-content/uploads/2024/07/civil-space-shortfall-ranking-july-2024.pdf>. [Accessed: Jun. 14, 2026].
- [3] NASA, “FY 2026–2027 SBIR/STTR Program Solicitation, Appendix 26B-I: SBIR Topics,” NASA Small Business Innovation Research Program, 2025.